

Mohd Bilal Khan

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EDUCATION

Indian Institute of Information Technology (IIIT), Pune

Bachelor of Technology in Electronics and Communication Engineering

Pune, India

Aug 2023 – May 2027

TECHNICAL SKILLS

Languages: C++, Python, Go (Golang), Java, JavaScript, Shell Scripting

AI/ML Frameworks: PyTorch, TensorFlow, LangGraph, RAG, CLIP, Transformers

Web & DevOps: React, Flask, Streamlit, Docker, Git, GitHub, Postman, Selenium

Databases: Neo4j (Graph DB), ChromaDB, FAISS, PostgreSQL, MongoDB

PROFESSIONAL EXPERIENCE

Student Researcher & Team Lead

Digital Forensics Lab, IIIT Pune

Oct 2025 – Present

Pune, India

- Leading a research team for a **4.2 Crore** government-funded project titled "Next Generation Data Carving and Preservation Techniques," in collaboration with NIT Raipur.
- Engineered an AI-powered **Vision Transformer (ViT)** module for violence detection, achieving **84.01% training accuracy** and surpassing baseline models by over **15%**.
- Developing a file tampering detection system utilizing **CLIP models** and Knowledge Graphs to automate evidence discovery for law enforcement.
- Expanding system architecture to support comprehensive multi-modal forensic data processing, encompassing text, audio, and video analysis.

KEY PROJECTS

EvoMind: Multi-Agent AI System | *LangGraph, RAG, Neo4j, Python*

- Architected a dual-agent state machine using **LangGraph** to intelligently route user queries to specialized therapist or logical agents based on structured message classification.
- Implemented real-time Knowledge Graph ingestion utilizing **Neo4j**, scaling to support over **10,000 nodes and relationships** via non-blocking pipelines.
- Designed a scalable system featuring a custom Web UI, Notion API synchronization, and speech I/O capabilities.

ZeroTrace: High-Performance Data Erasure Utility | *Go, Shell Scripting*

- Engineered a professional-grade security tool for Windows, Linux, macOS, and Android to perform irreversible data sanitization using **NIST SP 800-88** and DoD 5220.22-M protocols.
- Achieved a complete drive wipe on a **512GB NVMe SSD in 5.5 hours**, optimizing I/O operations for speed.
- Developed specifically for a national hackathon sponsored by the Government of India.

ACHIEVEMENTS & LEADERSHIP

National Hackathon Winner: Secured **Rank 130** (Top 1.3%) out of 10,000+ teams in the HackRx national hackathon.

Academic Excellence: Achieved a **Top 2% rank** (among 1.1 million candidates) in the JEE Mains examination.

Research Publication: Authored a manuscript (in preparation) on Chili Plant Disease Classification using CNNs, achieving **91% accuracy**.

Leadership: Selected as **Google DSC Campus Ambassador** for IIIT Pune; Active contributor to GirlScript Summer of Code (2024).

Athletics: Silver Medalist at the Inter-IIIT National Football Tournament.